Nelson–Siegel–Svensson Estimation of Greek and Italian Sovereign Discount Curves

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Abstract

A discount factor converts a future unit of currency into its time-zero present value. Under continuous compounding, the discount factor at maturity m is $D(m) = \exp[-my(m)]$, where $y(m)$ is the continuously compounded zero-coupon spot yield. This paper develops a reproducible framework for comparing Greek and Italian sovereign discount curves using the Nelson–Siegel–Svensson (NSS) model. The central methodological distinction is between coupon-bond yields-to-maturity, which summarize multiple cash flows, and zero-coupon spot rates, which directly determine discount factors. The repository workflow executes the empirical pipeline, validates the codebase, generates a results section directly from reproducible outputs, and compiles the resulting paper.

1 Introduction

A financial claim payable at a future date is not generally equivalent in present value to the same nominal claim payable today. For maturity m and continuously compounded spot yield $y(m)$,

$$D(m) = e^{-my(m)}. \tag{1}$$

The central empirical question is how to construct comparable model-implied discount curves from documented sovereign-bond observations.

2 Yield-to-Maturity and Zero-Coupon Curves

For a coupon-bearing government bond i,

$$P_i = \sum_{j=1}^{N_i-1} C_{ij} D(t_{ij}) + (C_{iN_i} + F_i) D(T_i). \tag{2}$$

Its yield-to-maturity compresses several cash-flow discount factors into a scalar internal rate of return. Therefore a benchmark yield-to-maturity cross-section is not, without additional assumptions, itself a directly observed zero-coupon discount curve.

3 Nelson–Siegel–Svensson Model

Let $\theta = (\beta_0, \beta_1, \beta_2, \beta_3, \tau_1, \tau_2)$. Define

$$\Lambda(\mu) = \frac{1 - e^{-\mu}}{\mu}. \tag{3}$$

The NSS spot-yield curve is

$$y(m;\theta) = \beta_0 + \beta_1 \Lambda\left(\frac{m}{\tau_1}\right) + \beta_2 \left[\Lambda\left(\frac{m}{\tau_1}\right) - e^{-m/\tau_1}\right] \tag{4}$$

$$+ \beta_3 \left[\Lambda\left(\frac{m}{\tau_2}\right) - e^{-m/\tau_2}\right], \tag{5}$$

with model-implied discount factor

$$D(m;\theta) = e^{-m y(m;\theta)}. \tag{6}$$

4 Two-Country Comparison

For $c \in \{\mathrm{GR}, \mathrm{IT}\}$,

$$\hat{\theta}_c = \arg\min_{\theta_c} \sum_{i=1}^{n_c} \left[\hat{y}_{i,c} - y(m_{i,c};\theta_c)\right]^2. \tag{7}$$

The bilateral comparison is

$$\Delta D(m) = \hat{D}_{\mathrm{GR}}(m) - \hat{D}_{\mathrm{IT}}(m). \tag{8}$$

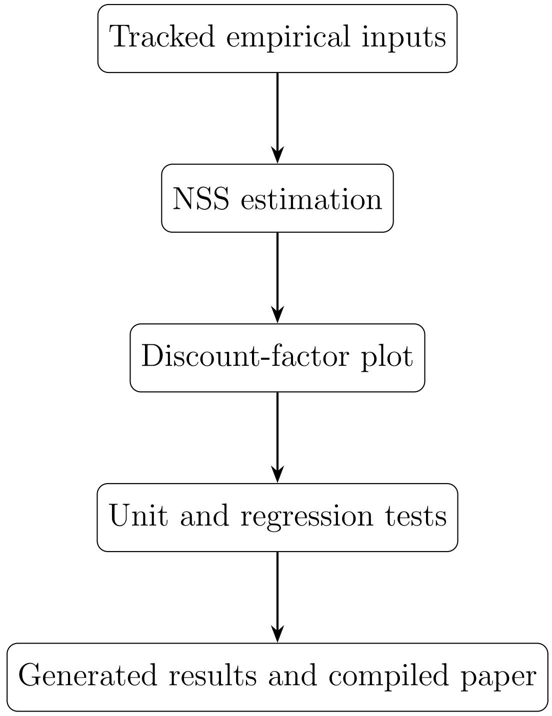
Under continuous compounding, for $m > 0$,

$$D_{\mathrm{GR}}(m) > D_{\mathrm{IT}}(m) \iff y_{\mathrm{GR}}(m) < y_{\mathrm{IT}}(m). \tag{9}$$

This follows from the strict monotonicity of the exponential function.

5 Reproducible Workflow

The computational architecture is



6 Reproducible June 2025 Greece–Italy Results

This section is generated automatically by the repository workflow from the empirical output files. The underlying inputs are benchmark yields transformed according to the repository's declared modeling convention; the resulting curves are therefore NSS model-implied discount factors rather than directly observed zero-coupon discount factors.

Table 1: NSS estimation diagnostics generated by the empirical pipeline.

Country	β_0	β_1	β_2	β_3	τ_1	τ_2	RMSE
Greece	0.35732110	-0.34046812	-0.14809965	-0.81407869	5.5000	25.5000	0.00004435
Italy	0.04604401	-0.02165477	-0.04818816	0.00020321	1.5000	8.0000	0.00000000

7 Interpretation and Limitations

The generated curves are NSS model outputs fitted under the repository's declared transformation convention. They must not be described as directly observed zero-coupon discount factors. Differences can reflect sovereign risk, inflation expectations, liquidity, maturity segmentation, monetary conditions, and estimation asymmetry. A market discount curve is also distinct from a social discount function used in welfare economics.

Table 2: NSS model-implied Greece–Italy discount-factor comparison.

Maturity	Greece	Italy	$\Delta D = D_{\mathrm{GR}} - D_{\mathrm{IT}}$
0.5	0.99121119	0.98946411	0.00174709
1.0	0.98164827	0.98037536	0.00127292
2.0	0.96006845	0.96041786	-0.00034941
3.0	0.93531750	0.93500818	0.00030932
5.0	0.87818206	0.87109869	0.00708337
7.0	0.81520694	0.80087065	0.01433629
10.0	0.72014715	0.69977250	0.02037466
15.0	0.58085573	0.55608047	0.02477526
20.0	0.46969468	0.44163295	0.02806173
30.0	0.29044017	0.27859393	0.01184624

8 Conclusion

The repository now treats reproducibility as an executable property: empirical inputs generate estimates, estimates generate the bilateral discount-factor comparison, tests validate the implementation, and successful validation generates the paper artifact. The central object remains

$$D(m) = e^{-my(m)}. \tag{10}$$

Glossary

Discount factor Present value of one unit payable at a future maturity.

Spot yield Continuously compounded yield on a zero-coupon claim.

Yield to maturity Internal rate of return summarizing a bond's contractual cash flows.

NSS model Nelson–Siegel–Svensson parametric representation of a term structure.

RMSE Root mean squared fitting error.

Reproducibility Ability to regenerate outputs from tracked inputs and declared procedures.

References

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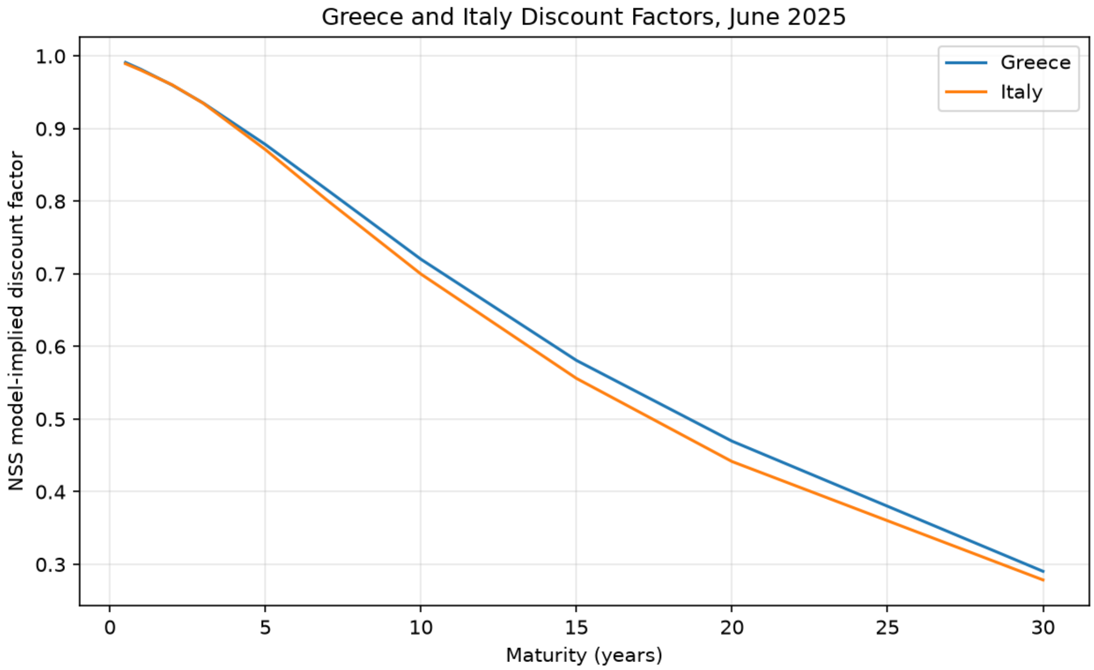

Figure 1: Automatically generated NSS model-implied discount-factor comparison.

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